161. Which of the following sulphides is yellow in colour

(a) CuS
(b) CdS
(c) ZnS
(d) CoS

Answer: B — (b) CdS

Chemical reason: CdS is characteristically yellow and is used as a yellow pigment. CuS and CoS are dark, while ZnS is white.

162. Which of the following is not oxidized by O_3

(a) KI
(b) $FeSO_4$
(c) $KMnO_4$
(d) K_2MnO_4

Answer: C — (c) $KMnO_4$

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in $KMnO_4$ has the oxidation-state change required by the stated redox conditions, so that choice is correct.

163. The number of moles of $KMnO_4$ reduced by one mole of KI in alkaline medium is

(a) One fifth
(b) Five
(c) One
(d) Two

Answer: D — (d) Two

Chemical reason: The n-factor is obtained from the oxidation-state change under the stated alkaline conditions; equivalent weight is molar mass divided by that n-factor.

164. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction

(a) $Na_2S_2O_3$ is oxidised
(b) CuI_2 is formed
(c) $Cu_2I_2$is formed
(d) Evolved I_2 is reduced

Answer: B — (b) CuI_2 is formed

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives CuI_2 is formed. The choice is consistent with the expected oxidation state and stability of the product.

165. The only cations present in a slightly acidic solution are Fe^{3+}, Zn^{2+} and Cu^{2+}. The reagent that when added in excess to this solution would identify and separate Fe^{3+} in one step is

(a) 2M HCl
(b) 6M NH_3
(c) 6M NaOH
(d) H_2S gas

Answer: D — (d) H_2S gas

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to H_2S gas. The choice is fixed by the known composition of that compound/mineral.

166. Which element is alloyed with copper to form bronze

(a) Fe
(b) Mn
(c) Sn
(d) Zn

Answer: C — (c) Sn

Chemical reason: Bronze is primarily an alloy of copper and tin, so tin is the element alloyed with copper to make bronze.

167. Emery consists of

(a) Impure corundum
(b) Impure carborundum
(c) Impure graphite
(d) Purest form of iron

Answer: A — (a) Impure corundum

Chemical reason: Emery is an impure form of corundum, Al_2O_3, containing iron oxides and other impurities; its hardness makes it a useful abrasive.

168. The metal commonly present in brass and german silver is

(a) Mg
(b) Zn
(c) C
(d) Al

Answer: B — (b) Zn

Chemical reason: Both brass (Cu–Zn) and German silver (Cu–Zn–Ni) contain zinc. Hence Zn is the metal common to the two alloys.

169. In the equation $4M + _8CN^- + _2H_2O + O_2$ $4[M(CN_2)]^- + _4OH^-$ The metal M is

(a) Copper
(b) Iron
(c) Gold
(d) Zinc

Answer: C — (c) Gold

Chemical reason: The correct choice is Gold. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

170. The term platforming is

(a) Platinum painting
(b) Flat sheet of platinum
(c) Platinum manufacturing
(d) Platinum used as a catalyst

Answer: D — (d) Platinum used as a catalyst

Chemical reason: Platforming is catalytic reforming of petroleum hydrocarbons over a platinum-containing catalyst; “plat-” refers to platinum used in the catalytic process.

171. Purple of cassium is

(a) Gold solution
(b) Silver solution
(c) Copper solution
(d) Platinum solution

Answer: A — (a) Gold solution

Chemical reason: Purple of Cassius is a colloidal gold preparation (gold associated with tin oxide) that gives a purple colour and is used in glass/ceramic coloration.

172. Match the items under List 1 with the compounds/elements from the List 2. Select the correct answer from the sets (a), (b), (c) and (d).

List 1 List 2 (i) Explosive (A) NaN_3

(ii) Artificial gem (B) Fe_3O_4 (iii)

Self reduction (C) Sn (iv)

Magnetic material (D) Al_2O_3

(e) $Pb(N_3)_2$ (F) Fe_2O_3 (G) Cu (H) SiC

(a) (i) A, (ii) D, (iii) G, (iv) B

(b) (i) A, (ii) D, (iii) G, (iv) F

(c) (i) E, (ii) D, (iii) G, (iv) B

(d) (i) E, (ii) H, (iii) C, (iv) F

Answer: C — (c) (i) E, (ii) D, (iii) G, (iv) B

Chemical reason: The correct choice is (i) E, (ii) D, (iii) G, (iv) B. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

173. Blood haemoglobin contains the metal

(a) Al

(b) Mg

(c) Cu

(d) Fe

Answer: D — (d) Fe

Chemical reason: Haemoglobin contains iron in the heme group. The Fe ion reversibly binds oxygen, enabling oxygen transport in blood.

174. Percentage of carbon in steel is

(a) 2.5 – 4.5%

(b) 0.25 – 0.5%

(c) 0.2 – 1.5%

(d) 3 – 3.5%

Answer: C — (c) 0.2 – 1.5%

Chemical reason: Steel contains an intermediate carbon percentage—roughly 0.1–2% depending on grade—between wrought iron and cast/pig iron. The option covering about 0.2–1.5% best matches the usual textbook range.

175. Steel is manufactured from

(a) Wrought iron

(b) Cast iron

(c)

(a) and

(b) both

(d) Haematite

Answer: C — (c) (a) and (b) both

Chemical reason: Steel is produced by controlling the carbon and impurity content of iron. Both cast iron and wrought iron can serve as starting materials in historical steelmaking routes.

176. Modern method for the manufacture of steel is

(a) Bessemer process

(b) Seimen-Martin's open hearth process

(c) Duplex method

(d) L.D. process

Answer: D — (d) L.D. process

Chemical reason: The LD (basic oxygen) process is a modern, rapid steelmaking method in which oxygen is blown through molten pig iron to oxidize excess carbon and impurities.

177. Spiegeleisen is an alloy of

(a) Fe, C and Mn
(b) Fe, Mg and C
(c) Fe, Co and Cr
(d) Fe, Cu and Ni

Answer: A — (a) Fe, C and Mn

Chemical reason: Spiegeleisen is a high-manganese pig-iron alloy containing Fe, C, and Mn. It was traditionally added during steelmaking to introduce manganese and carbon.

178. Stainless steel is an alloy steel of the following metals

(a) Fe Only
(b) Cr and Ni
(c) W and Cr
(d) Ni and Be

Answer: B — (b) Cr and Ni

Chemical reason: Common stainless steel contains chromium and nickel in addition to iron. Chromium gives passivation, while nickel improves toughness and corrosion resistance.

179. In the manufacture of steel, the Bessemer converter is containing lining of

(a) SiO_2
(b) CaO
(c) CaO and MgO
(d) Fe_2O_3

Answer: C — (c) CaO and MgO

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in CaO and MgO matches the relevant metallurgical process or alloy property.

180. Which of the following alloys contain only Cu and Zn

(a) Bronze
(b) Brass
(c) Gun metal
(d) Bell metal

Answer: B — (b) Brass

Chemical reason: Brass is the binary copper–zinc alloy. Bronze, gun metal, and bell metal contain tin as an important component.